

Glomerular Cell Biology and Podocytopathies

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KEY POINTS

- The glomerulus represents a functional and integrated syncytium of four types of glomerular cells, which together ascertain glomerular filtration.
- Together, podocytes and glomerular endothelial cells allow for a size- and charge-selective glomerular filtration due to their specialized three-dimensional structure, extensive glycocalyx coating, and the coordinated synthesis of the unique glomerular basement membrane.
- Mesangial cells provide structural support of the glomerular capillaries, regulate glomerular filtration, and maintain glomerular endothelial health.
- Parietal epithelial cells build the Bowman's capsule to prevent leakage of the primary urine filtrate to the tubulointerstitium, contribute to glomerular scarring, and are thought to constitute a potential reservoir for podocytes in development, maturation, and eventually in adulthood.
- The hallmarks of diseases related to primary podocyte injury of genetic or autoimmune origin are proteinuria (often nephrotic range), foot process effacement, podocyte hypertrophy, and depletion.
- Crosstalk between glomerular cell types attenuates and/or perpetuates glomerular injury.
- Most forms of glomerular injury result from humoral and cellular immunologic mechanisms to which the glomerulus reacts by basic responses such as cellular proliferation, changes in glomerular cell phenotypes, and increased deposition of extracellular matrix.

GLOMERULAR CELL ANATOMY AND INJURY RESPONSE PATTERNS

Loss of protein into the urine (proteinuria), especially albumin (albuminuria), is the hallmark of glomerular disease and an important prognostic marker for a wide variety of kidney diseases, including the numerically and economically increasing challenge of diabetic nephropathy.¹ Albuminuria is also an independent risk factor for cardiovascular mortality.² Therefore, understanding the pathophysiology of albuminuria and therapeutic approaches to its modification has major clinical and health economic significance.

The kidney of a healthy 70-kg adult filters approximately 180 L of plasma per day. Filtration and urine production

takes place in the smallest functional unit of the kidney, called the nephron. The ultimate site of filtration is located at the beginning of the nephron in the renal corpuscle. The renal corpuscle (Fig. 4.1) is composed of the glomerulus, a network (tuft) of capillaries, which is surrounded by parietal epithelial cells (PECs) of Bowman's capsule (BC), and is thereby separated from the tubular system. As described and depicted in detail in Chapter 3, blood enters the glomerulus at the vascular pole through an afferent arteriole of the renal circulation and is drained into an efferent arteriole. The unique resistance of these arterioles generates a high pressure within the capillary convolute of the glomerulus, ultimately driving the ultrafiltration of a primary urinary filtrate through the glomerular filtration barrier into Bowman's space with an effective filtration pressure of approximately 50 mm Hg.³

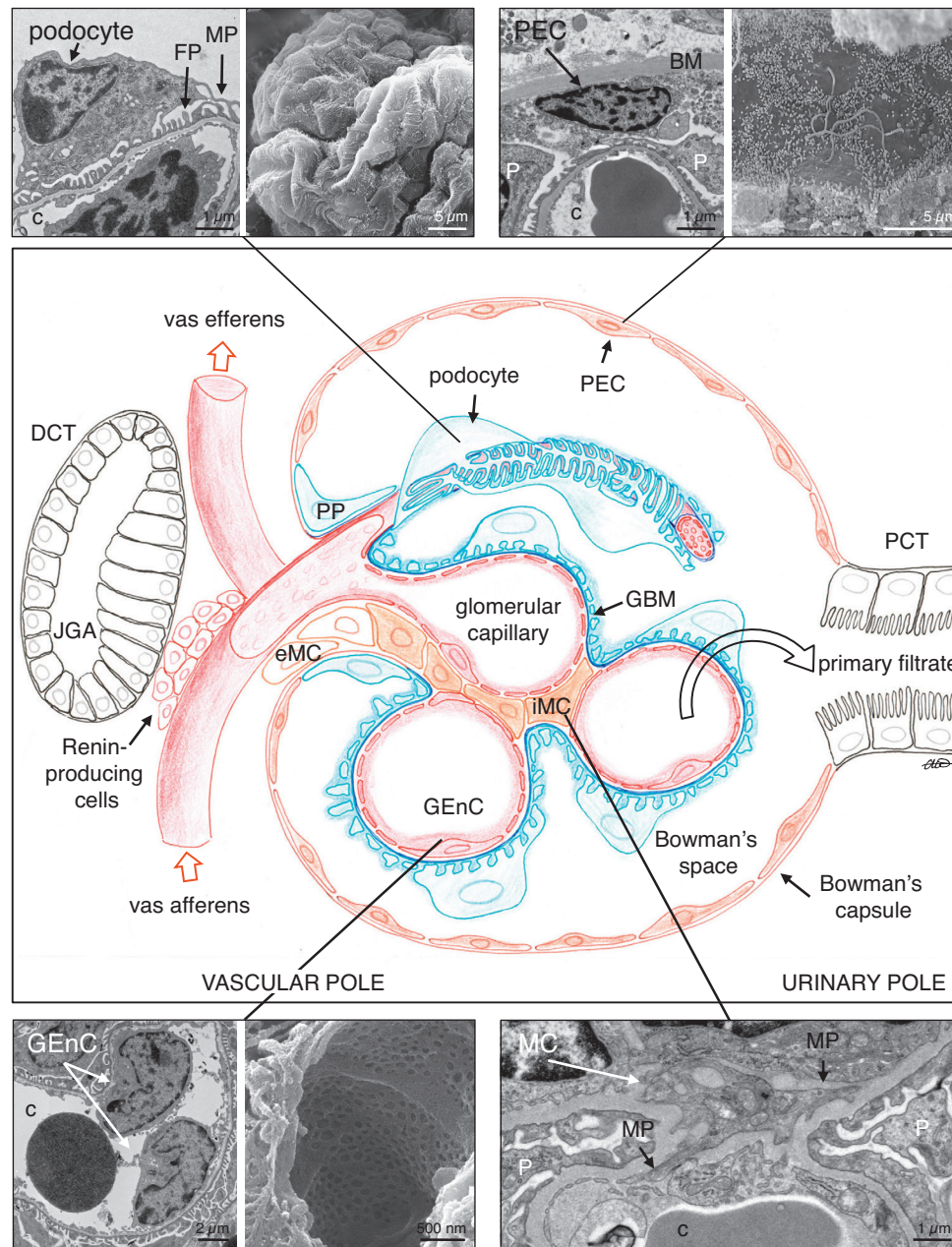


Fig. 4.1 Scheme of the glomerulus depicting the localization of the four resident glomerular cell types. The four types are podocytes, parietal epithelial cells (PEC), glomerular endothelial cells (GEnC), and mesangial cells (MC). Transmission and scanning electron micrographs show the typical ultrastructure of the glomerular cells. *BM*, Bowman's membrane; *c*, capillary lumen; *DCT*, distal convoluted tubule; *eMC*, extraglomerular mesangial cell; *FP*, foot process; *GBM*, glomerular basement membrane; *JGA*, juxtaglomerular apparatus; *iMC*, intraglomerular mesangial cell; *MP*, major process; *P*, podocyte; *PCT*, proximal convoluted tubule; *PP*, parietal podocyte. (All transmission electron microscopy and scanning electron microscope are from Oliver Kretz, Hamburg, Germany.)

Tracer studies point to a charge selectivity favoring the filtration of positively charged solutes.^{4,5} Furthermore, an electrokinetic model of filtration has been proposed. In this model, filtration pressure establishes a streaming potential across the glomerular filtration barrier with Bowman's space being more negatively charged than the endothelial lumen. This might establish a retrograde electrophoretic field that acts opposite to diffusive and convective fluxes and tends to exclude negatively charged macromolecules away from the glomerular filtration barrier in the course of active filtration.⁶ The primary urine exits the renal corpuscle at the urinary

pole to be further processed to final urine in the downstream tubular system. While water and small molecules such as glucose, salt, and amino acids freely pass across the glomerular filtration barrier, a partial impermeability to large molecules, such as albumin, into the primary urine is maintained. The degree of fractional albumin filtration has been a matter of debate, but more recent two-photon *in vivo* imaging studies indicate values for a glomerular sieving coefficient of 0.02–0.04 for albumin filtration into the primary urine, which for the most part is reabsorbed by the proximal tubule.⁷ This partial impermeability to large molecules is achieved by a highly

complex interplay of two types of glomerular cells: the visceral epithelial cells (podocytes) and the glomerular endothelial cells, which ultimately compose the three-layered glomerular filtration barrier (podocyte—the 250–400 nm thick glomerular basement membrane [GBM]—endothelial cell). The GBM is composed of several types of extracellular matrix (ECM) macromolecules (laminin, type IV collagen, the heparan sulfate proteoglycan agrin, and nidogen), which produce an interwoven meshwork thought to impart both size-selective and charge-selective properties.⁸ Finally, intraglomerular mesangial cells (MCs) occupy the space between the glomerular filtration barrier to provide structural support. As specialized pericytes, MCs indirectly participate in filtration by reducing the glomerular surface area by contraction and are also thought to participate in matrix turnover and innate immune function.⁹ Although all glomerular cells form a functional and integrated syncytium, in this chapter each component is described in brief.

Clinical Relevance

The glomerulus is a functional syncytium of four cell types, which interact in physiologic and also pathologic situations, thereby perpetuating and extending the site of glomerular injury.

PODOCYTES

STRUCTURE

Podocytes are highly differentiated, mesenchymal-derived cells.¹⁰ An apicobasal polarity axis allows for podocyte orientation between the urinary space and the GBM.¹¹ Podocytes reside in the urinary space and embrace the glomerular capillaries with their flat cell body, from which they extend long-branching major processes. Major processes give rise to secondary processes (often called foot processes), which interdigitate in a zipper-like fashion with secondary processes of neighboring podocytes¹² (Fig. 4.2). Foot processes attach podocytes to the underlying ECM of the GBM by specific proteins, such as adhesion receptors, including integrins, syndecans, vinculin, talin, and dystroglycan.¹³ Foot processes are interconnected by slit diaphragms, which represent highly sophisticated cell–cell contacts that form an adjustable, nonclogging barrier through which glomerular filtration occurs. Although the term “diaphragm” implicates a thin-layered sieve, recent studies revealed that the slit diaphragm is composed of multiple layers of flexible transmembrane molecules to limit the passage of macromolecules.¹⁴ Structurally, the slit diaphragm combines components of several types of cell–cell junctions, including tight, adhesion, neuronal, and gap junctions. Proteins from tight (ZO-1, JAM4, occludin, and cingulin) and adherens junctions (P-cadherin, FAT1, and the catenin family of proteins) are associated with the immunoglobulin superfamily members nephrin and neph1.¹⁵ Nephrin and neph1 form the core component of the slit diaphragm by a bipartite assembly with neph1 molecules spanning the lower part of the slit close to the GBM and nephrin molecules positioned in the apical side.¹⁴ The stomatin protein family member podocin (NPHS2) helps

anchor nephrin to the plasma membrane and generates a signaling hub in lipid rich membrane compartments, e.g., for the Ca^{2+} -permeable transient receptor potential channel (TRPC) 6¹⁶—that might translate mechanical tension to ion channel action and cytoskeletal regulation.¹⁷ On their intracellular C-terminal parts, nephrin and neph1 are associated with several signaling adaptor molecules and scaffold proteins linking the slit diaphragm to the actin cytoskeleton.^{14,18} In general, the podocyte cytoskeleton is highly elaborate. The podocyte cell body and major and secondary processes contain vimentin-rich intermediate filaments that assist in maintaining cell shape and rigidity. Large microtubules form organized structures along major processes. Foot processes contain long actin fiber bundles that run cortically and contiguously to link adjacent podocytes. Actin, α -actinin-4, and myosin form a contractile system in foot processes, which is regulated by the interplay of the actin binding proteins synaptopodin¹⁹ and α -actinin-4 with rho GTPases. This well-orchestrated actin and microtubule cytoskeleton ensures the high plasticity of the podocyte process network.^{20,21} The apical surfaces of podocytes are covered by the surface sialomucin podocalyxin,²² whose highly negative charge functions to keep adjacent foot processes separated, thereby keeping the urinary filtration barrier open.²³

FUNCTION

The main podocyte function is to build, maintain, and regulate the three-layered glomerular filtration barrier.

1. **The synthesis of the mature GBM** requires the crosstalk of podocytes with endothelial cells.²⁴ Podocytes and endothelial cells both secrete laminin 111 and type IV collagen $\alpha 1\alpha 2\alpha 1$ in GBM development and the final laminin 521 isoforms after maturation. However, only podocytes secrete type IV collagen $\alpha 3\alpha 4\alpha 5$ of the fully mature GBM.²⁵
2. **Podocytes maintain the glomerular filtration barrier** by secreting survival factors such as angiopoietin-1 (Angpt1, which binds to Tie2 on glomerular endothelial cells [GEnC] and MCs),²⁶ normosialylated angiopoietin-like-4 (which binds to integrin $\alpha V\beta 5$ on GEnC),²⁷ vascular endothelial growth factor A (binds to vascular endothelial growth factor receptor 2 [VEGFR2] on GEnC),²⁸ and stromal derived factor 1 (which binds to CXCR4 on GEnC)²⁹; these factors exert paracrine effects across the filtration barrier on glomerular endothelial and MCs supporting their respective migration, differentiation, and survival.³⁰
3. **Podocytes stabilize the glomerular filtration barrier** by expressing cell–matrix adhesion receptors such as integrin $\alpha 3\beta 1$, which connects laminin 521 in the GBM through various adaptor proteins to the intracellular actin cytoskeleton or integrins $\alpha 2\beta 1$ and $\alpha V\beta 3$, α -dystroglycan, syndecan-4, and type XVII collagen.³¹
4. **Podocytes regulate glomerular filtration**³² by presumably compressing the GBM through their adhesion to the GBM and the tensile forces of their cytoskeleton, which in turn reduces the permeability to macromolecules.³³ Furthermore, they regulate glomerular filtration through the formation of the slit diaphragm and by sensing the glomerular filtration pressure by a mechanoreceptor complex situated at the slit diaphragm.³⁴

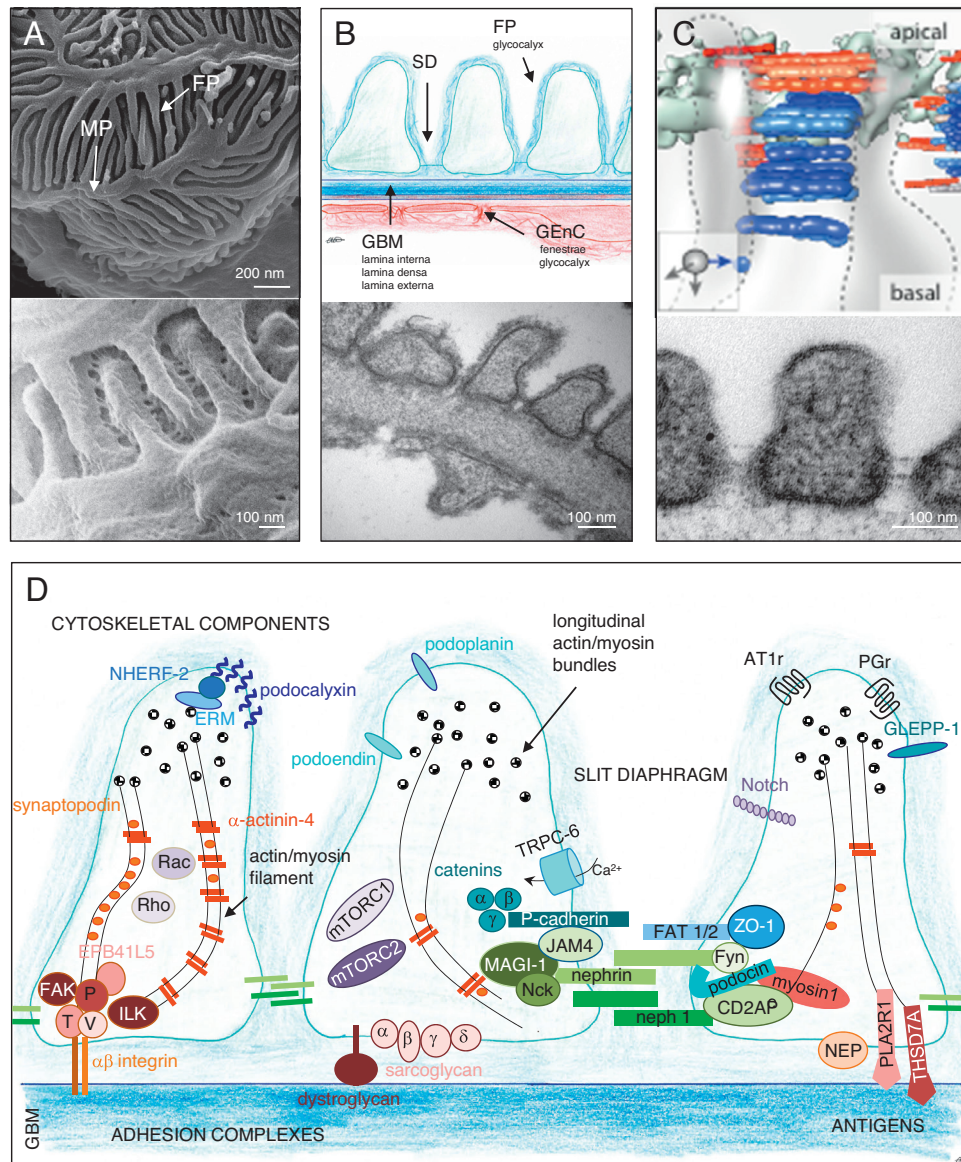


Fig. 4.2 Anatomy of the glomerular filter. (A) Scanning electron microscopy (SEM) of podocyte major processes (MP) and secondary interdigitating foot processes (FP). (B) Scheme and transmission electron microscopy (TEM) of three-layered glomerular filtration barrier consisting of (1) podocyte FPs covered by a glycocalyx, the specialized cell-cell contact called slit diaphragm (SD) connecting the interdigitating foot processes; (2) the three-layered glomerular basement membrane (GBM) consisting of podocyte-derived lamina interna, the podocyte/endothelial cell-derived lamina densa, and the endothelial-derived lamina externa; and (3) the glomerular endothelial cells (GEnC) with the large fenestrae and thick glycocalyx. (C) Scheme of the typical sagittal view of two FPs, showing three layers of ~25 nm long strands that are attributed to nephrin (blue, and 5 IG repeats) and one layer of ~40 nm long strands that represent nephrin (red, and 9 IG repeats). (D) Scheme of podocyte FP proteins constituting the (1) FP cytoskeleton (actin/myosin filaments, the actin binding proteins synaptopodin, α -actinin-4); (2) the podocyte adhesion complex ($\alpha\beta$ /1 integrin connecting the GBM to focal adhesions constituted of focal adhesion kinase [FAK], integrin linked kinase [ILK], talin [T], vinculin [V], paxillin [P], FERM domain protein EPB41L5); (3) matrix interacting proteins dystroglycan, sarcoglycan; (4) the slit diaphragm (with the transmembrane proteins nephrin, nephrin1, FAT1/2, P-cadherin, and the scaffolding protein ZO-1 [zonula occludens 1], the intracellular signaling hub constituted of podocin, TRPC6 [transient receptor potential cation channel, subfamily C, member 6], and the connection to the actin cytoskeleton through CD2AP, CD2-associated protein, the cytoskeletal adaptor protein Nck, the multidomain scaffolding protein MAGI-1 and the junctional cell adhesion protein JAM4); (5) the negatively-charged sialoprotein podocalyxin (which is connected to the plasma membrane by NHERF-1 = Na⁺/H⁺ exchanger regulatory factor 1 and ERM [ezrin/moesin/radixin proteins]) localizes to the surface of the plasma membrane, as do GLEPP-1 (glomerular epithelial cell membrane protein-tyrosine phosphatase 1), podoplanin, and podoendin; (6) the G-coupled receptors angiotensin receptor 1 (AT1r) and prostaglandin receptor (PGr); (7) the podocyte antigens neutral endopeptidase (NEP); phospholipase A2 receptor 1 (PLA₂R1), and thrombospondin type 1 domain containing 7A (THSD7A) identified in antenatal and adult membranous nephropathy; and (8) intracellular signaling molecules *Rho*, *Rac*, *mTORC1/2* (mammalian target of rapamycin complex 1 and 2), *Notch*, and *Fyn*. ([A, C] All TEM and SEM from mouse glomeruli are from Oliver Kretz, Hamburg, Germany. [B] TEM from mouse glomerulus exhibiting glycocalyx from K. Betteridge, K. Onions, and R. Foster, Bristol, UK. [C] Scheme from Grahmmer F, Wigge C, Schell C, et al: A flexible, multilayered protein scaffold maintains the slit in between glomerular podocytes. *JCI Insight* 1, 2016.¹⁴ [D] Drawn by C. Meyer-Schwesinger.)

PODOCYTE PATHOPHYSIOLOGY

Due to their molecular setup and localization, podocytes are permanently targeted by, and required to respond to, various physiologic and pathophysiologic stressors. If exposure is too excessive in time and dosage, this leads to complex adaptive and maladaptive intracellular changes, which then lead to the typical histopathologic sequence of foot process effacement, podocyte hypertrophy, and podocyte detachment from the GBM with loss into the urine (Fig. 4.3). Podocyte dysfunction results in clinical proteinuria and in a variety of glomerular responses, such as disruption of podocyte–endothelial crosstalk and activation of podocyte–parietal cell interactions culminating in glomerulosclerosis.

Over 80 pathways have been described that result in podocyte distress, including circulating factors, cell-surface signaling, metabolism, fibrosis, inflammation, and the actin

cytoskeleton.³⁵ As a general theme, podocyte injury appears to involve reactivation of developmental programs such as those engaged by Notch,³⁶ Wnt,^{37–40} and mTOR pathways.⁴¹ Overactivation, imbalance, and impairment of these central intracellular signaling pathways disrupt normal podocyte energy metabolism⁴² and protein homeostasis,⁴³ thus initiating a mostly irreversible dedifferentiation process.

Podocyte Foot Process Effacement

Podocyte function largely depends on its complex three-dimensional cytoskeletal structure. The identification of over 50 monogenic causes of podocyte disease⁴⁴ has immensely increased our understanding of podocyte function and dysfunction. Regardless of the underlying disease, a characteristic and almost predictable response to podocyte injury is a change in shape, called effacement.⁴⁵ Numerous studies have shown that effacement is an active process, due to

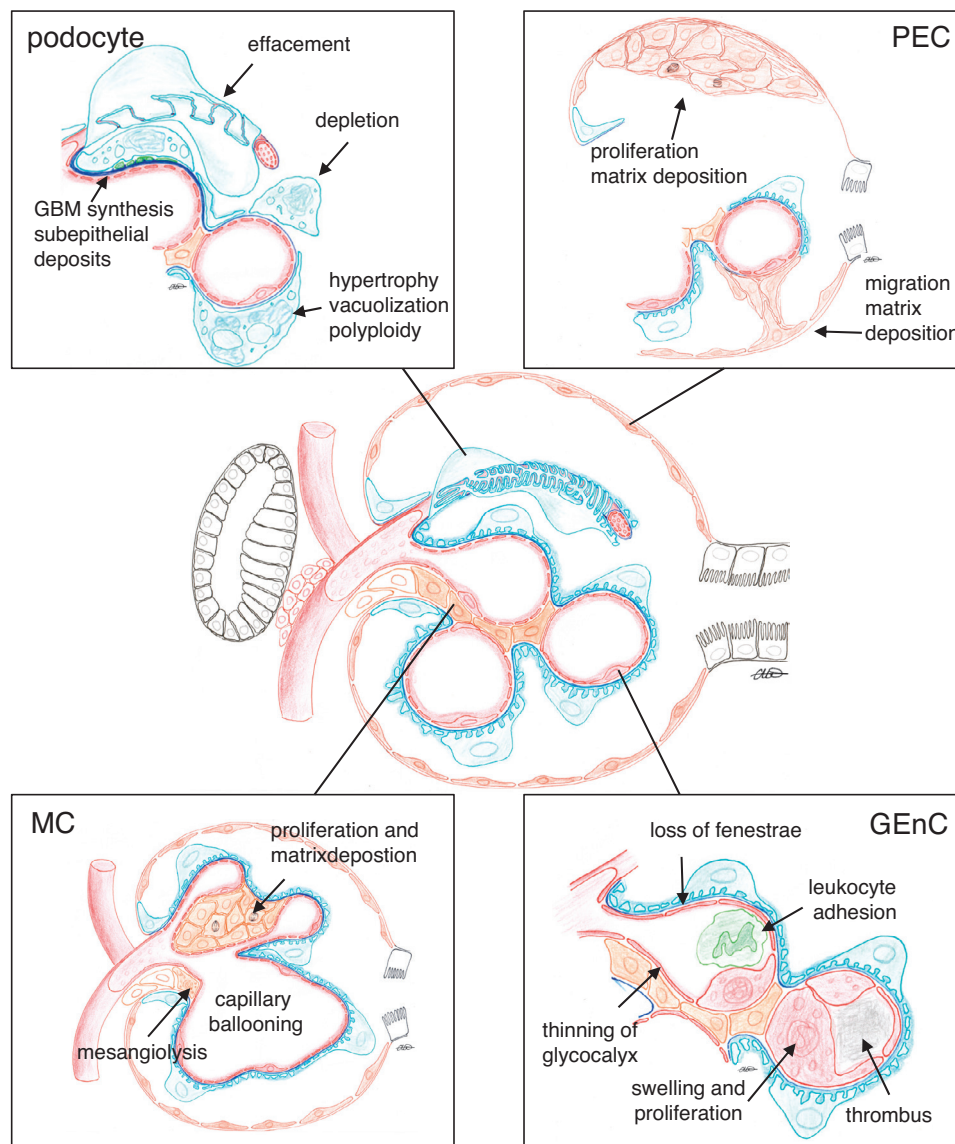


Fig. 4.3 Pathophysiological reaction patterns of glomerular cells. GENc, Glomerular endothelial cell; MC, mesangial cell; PEC, parietal epithelial cell.

changes in the actin cytoskeleton of the podocyte, which forms the “backbone” of these highly specialized cells.⁴⁶ Further evidence that effacement is an active process is that in some instances it can be reversed, such as in treatment-responsive patients with minimal change disease (MCD). There has been debate as to whether effacement per se causes proteinuria, because proteinuria due to podocyte damage can occur independent of this change in shape. The relationship between podocyte foot process effacement and proteinuria has been questioned,⁴⁷ and it is clear that there is still much to learn about this long-recognized but still poorly understood ultrastructural phenomenon. Confusingly, effacement has also been reported (in the absence of proteinuria) in the protein-malnutrition state kwashiorkor,⁴⁸ suggesting that it may be a feature of hypoalbuminemia rather than of proteinuria per se. However, it is generally accepted that effacement is a manifestation of serious podocyte injury, and that this histologic finding implies changes in either slit diaphragm proteins (i.e., nephrin⁴⁹ and podocin⁵⁰), actin binding and regulating proteins (i.e., α -actinin-4⁵¹ and CD2AF⁵²), podocyte attachment to the GBM (i.e., laminin β 2⁵³ and integrin β 4⁵⁴), nuclear proteins (WT1⁵⁵ and LMX1B⁵⁶), mitochondrial⁵⁷ and lysosomal⁵⁸ components, and/or other events, as genetic studies in humans suggest. Therefore teasing out precisely the biologic role of effacement in the development and maintenance of proteinuria may not be important.

Podocyte Hypertrophy

Podocytes are terminally differentiated epithelial cells and unable to adequately proliferate to cover denuded areas of the GBM in situations of glomerular (GBM) distention (i.e., in case of renal hyperfiltration) or podocyte loss. Despite the virtual absence of podocyte mitosis and regeneration, current knowledge suggests that differentiated podocytes do have at least some capacity to adjust to an altered glomerular architecture by hypertrophy. Therefore hypertrophy can be adaptive in the setting of glomerular development, growth, and numerical minimal podocyte depletion (up to 20%), or can reflect a multifactorial maladaptive response of podocytes due to persistent injury promoting stimuli such as high glucose in diabetic nephropathy or subepithelial deposits in membranous nephropathy.⁵⁹ Recent findings highlight mammalian target of rapamycin (mTOR) and its downstream target, the translational repressor protein 4E-BP1,⁶⁰ as a key regulator of both adaptive and maladaptive podocyte hypertrophy, whereby the timing, extent, and duration of mTOR activation decide whether hypertrophy is adaptive or maladaptive.⁴¹ Inhibition of mTOR by rapamycin in the setting of adaptive hypertrophy results in proteinuria and glomerulosclerosis, whereas inhibition of mTOR in the setting of maladaptive hypertrophy could be therapeutically beneficial.⁶¹ Besides an imbalance of mTOR signaling pathways,^{62,63} podocyte hypertrophy in response to hyperglycemia and stretch have been shown to be also mediated by the cyclin-dependent kinase inhibitor p27Kip1.^{64,65} Hypertrophy in membranous nephropathy seems to originate in part from altered protein degradation and subsequent cytoplasmic accumulation of proteins.⁶⁶

Podocyte Depletion

Many podocyte diseases such as focal segmental glomerulosclerosis (FSGS), membranous nephropathy (MN), and

diabetic kidney disease are accompanied by a progressive decline in overall kidney function, measured clinically by a decrease in glomerular filtration rate. This is largely due to glomerulosclerosis, with or without tubulointerstitial fibrosis. Patterns of glomerulosclerosis histologically include a segmental form (a portion of an individual glomerulus is scarred) and the more extensive global form (the majority of an individual glomerulus scars). Podocyte depletion is a major contributor to the development of age-related glomerulosclerosis in humans and rodents.^{67–70} A decrease in podocyte number is one of the best predictors for a poor outcome in clinical diabetic kidney disease. A loss of 20% of podocytes is tolerated by rats⁷¹ and mice⁷² and is accompanied by MC proliferation and expansion. Segmental glomerulosclerosis ensues when 40% of podocytes are depleted, and global glomerulosclerosis occurs when the podocyte number is below 60% of normal.⁷¹ There has been a long-standing debate on the underlying mechanisms for podocyte depletion, ranging from necrosis, apoptosis, and necroptosis to the detachment of viable cells from the GBM.⁷³ Interestingly, viable podocytes can be isolated from the urine of proteinuric patients, emphasizing the importance of podocyte detachment as a mechanism of podocyte depletion in glomerular diseases.

Studies have suggested that despite a lack of proliferation, podocyte number can be restored following certain therapies such as angiotensin-converting enzyme inhibition.⁷⁴ Despite compelling data, however, it remains unclear whether podocyte regeneration exists in renal aging or in pathophysiologic situations and, if podocyte regeneration exists, from what source of resident progenitor cells the novel podocytes originate.⁷⁵ Possible sources discussed might be progenitor cells derived from glomerular parietal epithelial cells⁷⁶ and/or cells of renin lineage,⁷⁷ although further studies are needed to fully validate these findings.

Podocyte-Related Mechanisms of Proteinuria

Proteinuria is the clinical hallmark of podocyte injury, ensuing from a disruption of their most important biologic function, which is namely to limit the passage of plasma proteins into the urinary space. In general, any kind of injury affecting podocyte function results in proteinuria, either selective as albuminuria (mostly loss of the 60-kDa protein albumin) or nonselective as global proteinuria (loss of a multitude of proteins over 60 kDa of size, including immunoglobulins of 150 kDa) in case of major breakdown of the glomerular filtration barrier. Genetic studies and animal studies have demonstrated that podocyte-dependent proteinuria originates from at least six distinct mechanisms.

Mechanism 1: Alteration of the slit membrane as the size- and charge-selective barrier to proteins through hereditary or acquired defects of one or more structural slit diaphragm proteins, such as nephrin or podocin, leads to increased passage of proteins across this barrier. Either an absolute decrease in slit diaphragm protein levels or a change in their subcellular location is associated with proteinuria. Moreover, given the complex interplay between proteins comprising the slit diaphragm, a change in one slit diaphragm protein often leads to a cascading dysfunction of one or more of the other proteins.⁷⁸

Mechanism 2: Alteration of the podocyte cytoskeletal network either through mutations of structural proteins and

adaptor proteins or through an imbalance of signaling through actin-regulating enzymes and of converging signaling pathways culminate in enhanced or decreased actin polymerization, ultimately leading to proteinuria through a loss of the complex three-dimensional structure and flexibility of the podocyte cytoskeleton.

Mechanism 3: Although no known hereditary mutations of podocalyxin have been described, loss of podocalyxin results in reduced negative surface charge of podocyte processes and proteinuria.²³

Mechanism 4: Podocyte depletion (i.e., through loss of adhesion)⁷⁹ results in denuded GBM areas through which proteins escape.

Mechanism 5: Alteration in GBM composition by podocytes through increased secretion of ECM proteins is typically observed in membranous and diabetic nephropathy. These ECM proteins are laid down along the GBM and eventually lead to the characteristic thickening of the GBM in these diseases.⁸⁰ The altered ECM composition leads to secondary changes (loss) in negative charge to the GBM, thereby enabling increased passage of proteins. In addition, increased production and secretion of reactive oxygen species and metalloproteinases from podocytes leads to degradation of the GBM and proteinuria.

Mechanism 6: Effect on the glomerular endothelial cell whose survival is dependent in part on the VEGFA produced and secreted from podocytes.⁸¹ A decrease in production by podocytes, such as when podocyte number decreases, leads to secondary apoptosis of glomerular endothelial cells, which in turn is accompanied by a decrease in resistance of this layer of the glomerular filtration barrier.

MESANGIAL CELLS

STRUCTURE

MCs are divided into intraglomerular and extraglomerular MCs. MCs are mainly derived from the metanephric mesenchyme⁸² and migrate into the cleft of the comma and S-shaped bodies as well as the maturing glomerulus under the chemotactic control of platelet-derived growth factor B⁸³ and the survival factor VEGF, both secreted from the progenitors of podocytes and GEnC.⁸⁴ In the mature glomerulus, MCs constitute the central stalk and are in continuity with the extraglomerular mesangium and the juxtaglomerular apparatus. Extraglomerular MCs are in close connection to afferent and efferent arteriolar cells by gap junctions, allowing for intercellular communication.⁸⁵ MCs are highly branched with processes extending in all directions. First, MCs have major processes that contain abundant bundles of microfilaments, microtubules, and intermediate filaments.⁸⁶ These major processes contain actin, myosin and α -actinin, and connect MCs with anchoring filaments to the GBM opposite podocyte foot processes and at paramesangial angles, giving them contractile properties.⁸⁷ Second, MCs have abundant microvillus-like projections arising from the cell body or from the major processes. Within the cell body microfilament bundles are less frequent and the perinuclear region is free of microfilaments.⁸⁶ MCs are in direct contact with GEnC without an intervening basement membrane on the capillary lumen side, where the cell membranes of both cell types interdigitate.⁸⁸ MCs do not express highly cell-specific markers; however, they express genes that classify MCs as a

special form of microvascular pericytes.⁸⁹ MCs are positive for vimentin and desmin.

FUNCTION

MCs, together with their matrix, form a functional unit with GEnC and podocytes.⁹

1. MCs are required for the development and for the mechanical (structural) support of glomerular capillary loops.^{90,91} Mechanical support is in part mediated by attachment of MCs to the carboxy-terminus globular domain of laminin $\alpha 5$ in the GBM.⁹²
2. MCs regulate the glomerular capillary flow and ultrafiltration surface and hence the fine-tuning of glomerular filtration by cell contraction at the single nephron level.⁹³ Contraction of MCs is regulated by vasoactive substances⁹⁴ and is dependent on the calcium signaling response and membrane permeability.^{94,95} Relaxation of MCs, on the other hand, is mediated by paracrine factors, hormones, and cyclic adenosine monophosphate (cAMP), with a role for growth factor priming.⁹⁴
3. MCs are responsible for the homeostasis of the mesangial matrix through synthesis and degradation of their own matrix components (type IV collagen $\alpha 1\alpha 2$, type V collagen, laminin fibronectin, proteoglycans [heparan and chondroitin sulfate, decorin, and biglycan], entactin, and nidogen).^{83,96-98} The mesangial matrix provides structural support to the glomerulus⁹ and regulates the behavior of MCs, such as growth and proliferation,⁹⁹ partly by binding and sequestering growth factors and thereby influencing their activation and release.⁹ Furthermore, the mesangial matrix signals to MCs in response to mechanical stretch.¹⁰⁰
4. MCs are sources and targets of growth factors, cytokines, and vasoactive agents.⁹ For example, MCs produce transforming growth factor (TGF)-1, VEGF, and CTGF⁹ in response to capillary stretching resulting from glomerular hypertension. MC proliferation, along with eicosanoid and matrix production, is influenced by PDGF-B,¹⁰¹ PDGF-C,¹⁰² fibroblast growth factor (FGF),⁹⁶ hepatocyte growth factor (HGF),⁹⁷ connective growth factor (CTGF),¹⁰³ epidermal growth factor (EGF),¹⁰⁴ and TGF- β .⁹⁸
5. MCs keep the mesangial space free from accumulating macromolecules, which trespass from the capillary lumen through the fenestrated endothelium. For this purpose, MCs phagocytose glomerular basal lamina or immune complexes formed at or delivered to the glomerular capillaries,^{105,106} and aid neutrophils to phagocytose apoptotic cells.¹⁰⁷ Macromolecules are removed by both receptor-dependent and independent mechanisms, depending on the size, charge, concentration, and affinity for MC receptors.^{9,108}
6. MCs are involved in the tubuloglomerular feedback by communicating with vascular smooth muscle cells over gap junctions.¹⁰⁹
7. MCs maintain endothelial health and function by cross communication via the mediators and pathways described above.

PATHOPHYSIOLOGY

As of yet, no discrete primary disease of MCs has been described. However, MCs react to changes of the intravascular milieu (soluble factors), immunoglobulin deposition, and

to changes affecting GEnC and podocytes. Glomerular MC injury is commonly associated with mesangial immune deposit formation in IgA nephropathy, lupus nephritis, and Henoch-Schönlein purpura (also designated as IgA vasculitis).¹¹⁰ Even though IgA nephropathy primarily affects the mesangium (mesangioproliferative disease), it produces marked hematuria and proteinuria, indicating changes in the permeability of endothelial cells, GBM, and podocytes. The myriad biologic responses of MCs to injury range from mesangiolysis to mesangial hypercellularity, mesangial expansion, and the promotion of glomerular inflammation (see Fig. 4.3). Overall, these biologic responses of MCs to injury are elicited (1) by structural abnormalities of the basement membrane that are either acquired (as in diabetes mellitus) or genetic (as in Alport syndrome); (2) by factors either released from neighboring MCs in an autocrine manner; or (3) from circulating factors affecting MCs in an endocrine manner.

Mesangiolysis is defined as a dissolution or attenuation of mesangial matrix and degeneration of MCs either by apoptosis or lysis without obvious damage to the capillary basement membranes. The matrix swells, loosens, and eventually dissolves; the MCs may show swelling and vacuolization. Mesangiolysis results in a dilation of the glomerular capillary lumina, as the mechanic support of capillaries provided by the anchoring points of MCs to the GBM is lost.¹¹¹ Loss of intraglomerular MCs can be replenished by ingrowth of extraglomerular MCs.¹¹²

Mesangial hypercellularity is characterized by an increase of intraglomerular MC number by hypertrophy, proliferation, and migration⁸² as in IgA nephropathy. In cases where the proliferative insult is limited, mesangial hypercellularity is limited by apoptosis and phagocytosis of these apoptotic cells by adjacent MCs or by infiltrating inflammatory cells.¹¹³

Mesangial expansion is a term characterizing the widening of the intraglomerular mesangium. Mesangial expansion occurs in diabetic nephropathy by excess mesangial matrix production such as fibronectin by MCs¹¹⁴ or by decreased degradation of the mesangial matrix by metalloproteinases.¹¹⁵ Mesangial expansion also occurs through the deposition of immune complexes, light chains, amyloid, fibrils, complement, and worn-out GBM material.¹¹⁶

Promotion of glomerular inflammation: Injured MCs generate reactive oxygen species, proinflammatory activators such as platelet activating factor,¹¹⁷ cytokines (TNF- α , CSF-1 and IL-6), and chemokines,⁹ thus sustaining and perpetuating glomerular inflammation.

GLOMERULAR ENDOTHELIAL CELLS

STRUCTURE

The glomerular microcirculation is unique as unlike other capillary circulations, glomerular capillaries are highly permeable to water and small solutes while maintaining relative impermeability to macromolecules potentially even as small as albumin.¹¹⁸ GEnCs are highly specialized cells that form the continuous inner layer of glomerular capillaries. GEnCs have a particular embryonic origin, with the majority of GEnCs arising by vasculogenesis from mesenchymal precursors in combination with a minority of GEnCs arising from introgession of existing vessels.^{119,120} In the mature glomerulus, the nucleus of GEnCs bulges into the capillary lumen. The

cytoplasm of GEnCs is 200 nm thin at its slimmest areas and punctuated by numerous fenestrae. The fenestrae of GEnCs are the largest in comparison to the fenestrae of endothelial cells of other organs and represent circular pores of 60 to 80 nm in diameter, which cover 20% of the glomerular endothelial surface.¹²¹ Under special fixation conditions, a diaphragm is visible that spans the fenestrae.¹²² GEnCs are covered by a 200- to 400-nm-thick glycocalyx, which represents a negatively charged gel-like surface structure of proteoglycans with their covalently bound polysaccharide chains named glycosaminoglycans (GAGs), glycoproteins, and glycolipids. The main carbohydrate constituents are heparan sulfate (HS), chondroitin sulfate (CS), and hyaluronan (HA), bound to the HA binding surface proteins such as CD44. The glycocalyx is attached to the GEnC by charge-charge interactions,¹²³ rendering GEnC very sensitive to hemodynamic factors¹¹⁸ such as shear stress.¹²⁴ The glycocalyx covers the fenestrae and the interfenestral domains of GEnC equally¹²⁵; however, the thickness of the glycocalyx differs between fenestrated and nonfenestrated GEnCs.¹²⁶ The GEnC fenestrae are plugged by a high content of HA¹²⁶ and are considered to be a key component of the glomerular permeability barrier. Like other endothelial cells, GEnCs express the specific markers platelet endothelial cell adhesion molecule 1 (PECAM1, CD31), intercellular adhesion molecule 2 (ICAM2), VEGFR2, growth factor receptor Tie2, von Willebrand factor, and vascular endothelial (VE)-cadherin (CD144).¹²⁷

FUNCTION

1. GEnCs are involved in the GBM production together with podocytes but to a lesser extent than podocytes as seen in mice with podocyte-specific type IV collagen $\alpha 5$ chain deletion (Alport model), which exhibit marked thinning and alteration of the GBM.^{128,129}
2. GEnCs contribute to the hydraulic conductivity of the glomerular filtration barrier through GEnC fenestrations,¹³⁰ which are formed in response to VEGF.^{127,131}
3. GEnCs contribute to the size and charge selectivity of the glomerular filtration barrier by the endothelial surface lining composed of the membrane-bound glycocalyx and the loosely bound endothelial cell coat.^{126,132} The glycocalyx adds to size and charge selectivity, most likely by forming a mesh-like structure of negatively charged HSs and the less charged HA,¹³³ as infusion of enzymes that degrade the glycocalyx increases albumin passage through the GEnC.¹³⁴
4. The glycocalyx of GEnC protects against protein leakage, inflammation, and coagulation.¹³⁴ With properties like a hydrogel, the glycocalyx acts as a size barrier to protein. With the high negative charge of polyanions, the glycocalyx electrically repels proteins.^{135–137} GAG-degrading enzymes such as chondroitinase and heparanase alter glomerular permeability.^{138,139} The gel-like, antiadhesive properties of the glycocalyx preclude the interaction of leukocytes with adhesion molecules.

PATHOPHYSIOLOGY

GEnCs are primarily targeted in several forms of vasculitis, in hemolytic uremic syndrome, and in preeclampsia.¹⁴⁰ Even though GEnCs are primarily affected, these conditions are associated with mesangiolysis and proteinuria, indicating endothelium-dependent changes in MCs and podocytes. Injury

of the GEnC induces the release of vasoactive substances and changes the composition of the endothelial glycocalyx and endothelial adhesion molecules, resulting in a net prothrombotic state (see Fig. 4.3).^{141,142} Furthermore, GEnCs hypertrophy, proliferate, go into apoptosis, and detach, further accelerating the thrombosis of glomerular capillaries.¹⁴¹ Experimental models indicate that recovery from glomerular injury is dependent on GEnC angiogenesis.^{143–145} Visualization of GEnC injury remains challenging. Morphologic signs of GEnC injury are swelling of the cell body, thinning of the glycocalyx,^{146,147} loss of fenestrations, and enhanced expression of adhesion markers such as CD34¹⁴⁸ and E-selectin (CD62E), an adhesion receptor for leukocytes. GEnC injury has been demonstrated to arise from an altered crosstalk with podocytes.

Clinical Relevance

Proteinuria is a hallmark of glomerular injury and should usually initiate a workup by a nephrologist to identify eventual underlying genetic, inflammatory, toxic, tumor, and infectious causes.

Endotheliosis: Glomerular capillary endotheliosis describes the swelling of GEnC with the deposition of fibrous material in and beneath GEnC, a condition typically seen in pre-eclamptic glomerular injury. The net result of GEnC swelling and deposition of fibrous material is capillary occlusion.

Changes of glycocalyx and their sequelae because the glycocalyx functions as a molecular scaffold and binds (1) circulating proteins such as growth factors and chemokines; (2) proteins involved in cell attachment, migration, and differentiation; and (3) proteins involved in blood coagulation and inflammation, alterations of the glycocalyx are central to the pathology of GEnC and glomerular injury. GEnC injury induces changes both in the thickness and in the molecular composition of the endothelial glycocalyx. Changes in glycocalyx thickness are mainly due to upregulation of glycocalyx-degrading enzymes such as hyaluronidase, heparanase, and proteinases, thereby shedding glycocalyx fragments into the glomerular circulation. Alteration of glycocalyx composition such as decreased levels of HA or changes in HS sulfation patterns change the antiadhesive and anticoagulative properties of the glycocalyx. In total, loss of glycocalyx thickness and glycocalyx modification result in enhanced permeability of the filtration barrier to proteins, inflammation, and coagulation.¹³⁴

Mediation of the inflammatory reaction: Injured GEnCs release vasoactive substances (nitric oxide and endothelin), which regulate glomerular filtration. Furthermore, GEnCs promote glomerular inflammation by attracting leukocytes by means of shed glycocalyx fragments and by expressing leukocyte adhesion molecules. Upon perturbation of the glycocalyx, these adhesion molecules become unmasked and allow leukocyte interactions with the endothelial surface. HS of the glycocalyx acts as a direct ligand for L-selectin.¹⁴⁹ Upon stimulation with inflammatory stimuli, GEnCs increase the expression of HS domains,¹⁵⁰ which facilitates leukocyte extravasation.¹⁵¹ The endothelium binds chemokines, which

regulate the extravasation of leukocytes upon a chemoattractant gradient.¹⁵² Alterations of glycocalyx composition ensue in enhanced binding of chemokines by way of positive charge interactions.¹⁵³

PARIETAL EPITHELIAL CELLS

PECs are derived from the metanephric mesenchyme and line Bowman's capsule of the renal glomerulus. During the vesicle and comma stages in glomerular development, PECs share a common phenotype with the other epithelial cells of the later glomerulus—namely, podocytes and proximal tubular cells. With the formation of the Bowman's space in the S-shaped body, the phenotypes of PECs, podocytes, and proximal tubular cells diverge. In the mature kidney, PECs are a heterogeneous population of cells as PECs at the urinary pole maintain features of proximal tubular cells and PECs at the vascular pole maintain features of podocytes and are therefore termed parietal podocytes.¹⁵⁴

STRUCTURE

PECs resemble squamous epithelial cells with their small thin cell body ranging in thickness from 0.1 to 0.3 μm and increasing to 2.0 to 3.5 μm at the nucleus. The surface of some PECs is lined by microvilli and cilia in a range from zero to two cilia per cell.¹⁵⁵ They are interconnected by “labyrinth-like” delicate tight junctions located at the apical surface, which consist of claudin-1, -2, -3, K-cadherin (Cdh6), kidney specific cadherin (Cdh16), occludin, and zonula-occludens 1 (ZO-1).¹⁵⁶ PECs have a simple cytoskeleton with filaments at the basal membrane region.¹⁵⁷ They express the intermediate filament protein cytokeratin 8.¹⁵⁸ PECs have the transcriptional prerequisite to express podocyte markers.¹⁵⁹ The expression of podocyte proteins is negatively regulated through protein degradation¹⁵⁹ and by microRNA-193a,¹⁶⁰ which represses Wilms tumor protein 1 (WT1) mRNA levels.¹⁶¹ PECs express the transcription factor Pax2 from the paired box family, which is involved in regulating genes governing proliferation, cell growth, and survival. Furthermore, PECs can be differentiated from podocytes through the expression of EPH receptor A7 belonging to the ephrin receptor subfamily, ladinin (a proposed anchoring filament that is a component of basement membranes), and scinderin (a calcium-dependent protein that regulates cortical actin networks).¹⁶²

FUNCTION

1. PECs at the vascular pole constitute a potential reservoir for podocytes in glomerular development, maturation,^{72,163} and eventually even adulthood.¹⁶⁴
2. PECs presumably form the basement membrane of Bowman's capsule, which consists of laminin-111, laminin-511,^{165,166} type IV collagen $\alpha 1 \alpha 2$, and type IV collagen $\alpha 5 \alpha 6$ ^{167,168}; however, this is not definitely proven.
3. PECs prevent the leakage of urine from the urinary space into the periglomerular compartment.¹⁶⁹

PATHOPHYSIOLOGY

To date there is no evidence for glomerular injuries related to primary PEC injury. However, PECs have taken the center stage of attention for their contribution to glomerular diseases such as rapid progressive glomerulonephritis (RPGN) and

FSGS. Activated PECs exhibit a larger cytoplasm and larger rounder nuclei. Further signs of PEC activation are a *de novo* expression of CD44¹⁷⁰ and the phosphorylation of signaling molecules.^{171,172} The predominant reactions of PECs to glomerular injury are proliferation, migration, and synthesis of a matrix that results in the development of crescents or glomerular tuft scars (see Fig. 4.3). There is an ongoing debate whether PECs serve as an intrinsic fixed progenitor population to replenish podocytes or proximal tubular progenitor cells in glomerular regeneration.¹⁷³ PECs respond to injury with the expression of podocyte markers such as synaptopodin and WT1.^{174,175}

Proliferation and migration: Activation of PECs as in rapid progressive glomerulonephritis (RPGN) results in their proliferation and the formation of extracapillary proliferations, also called cellular glomerular crescents, of which PECs are the major constituents.¹⁷⁶ Crescent formation is not only the result of PEC proliferation but also in part the result of the transdifferentiation of PECs to myofibroblasts,^{170,177} the deposition of matrix, and the infiltration with inflammatory cells. Crescents can occlude the tubular outlet of the glomerulus, resulting in entire nephron degeneration,¹⁷⁸ and are generally associated with a poor prognosis. PEC proliferation is usually associated with glomerular endothelial cell, GBM, or podocyte injury, as leakage of plasma components such as fibrin from the blood circulation is a strong inducer of PEC proliferation.^{179,180} Additionally, PEC depletion initiates PEC proliferation and crescent formation.¹⁸¹ Migration of PECs from Bowman's capsule onto the glomerular tuft is the predominant reaction of PECs in focal segmental glomerulosclerosis (FSGS).¹⁷⁶ In this scenario, PECs are activated and invade the glomerulus at focal areas via adhesions connecting sclerotic capillary areas with Bowman's capsule.¹⁸² PECs are then present in the sclerotic regions^{183,184} and can be visualized by staining for CD44.¹⁸⁵

Matrix deposition: Thickening of Bowman's capsule due to matrix production can be observed in aging glomeruli¹⁸⁶ and glomerular injury. Matrix deposition by PECs is observed in glomerular crescents and in the sclerotic glomerular tuft regions in FSGS. PECs that migrate to the glomerular tuft produce and deposit ECM proteins. The composition of this ECM is related to that of Bowman's capsule¹⁸³ and contains specific heparan sulfate moieties of heparan sulfate proteoglycans.¹⁸⁷

GLOMERULAR CELL CROSSTALK

Podocytes, glomerular endothelial cells, MCs, and PECs with their respective matrix must be considered as a functional unit, in which every cell plays its part in ensuring proper glomerular filtration. As a consequence, 30 years of isolated cell-type-based research is now being replaced by systems biology approaches to integrate the role and contribution of each individual glomerular cell type for the proper glomerular biology and function. Recent advances have demonstrated that glomerular cell crosstalk is the prerequisite for normal glomerular development and health. Furthermore, primary injury of one glomerular cell type affects the other glomerular cell types by crosstalk. Clinical and experimental observations suggest that crosstalk exists between podocytes and GEnC, between GEnC and MCs, between MCs and podocytes, and finally between podocytes and PECs (Fig. 4.4).

PODOCYTES AND GLOMERULAR ENDOTHELIAL CELLS

Podocytes are essential for GEnC development and maintenance, and the first glomerular cell crosstalk to be identified was the cross communication from podocytes to GEnC. Podocytes secrete vascular growth factors, such as VEGFA, which bind to its cognate receptor VEGFR2 on GEnC, a crosstalk crucial for GEnC health⁸¹ and which, if disrupted, results in glomerular injury. A tight control of VEGFA levels is essential for correct glomerular barrier function, and even though other podocyte-specific proteins such as the transcription factor Pod1/Tcf21¹⁸⁸ or the TGF- β activated kinase 1 (Tak1)¹⁸⁹ have been shown to be essential for GEnC development and health, it remains to be established whether this is a direct effect or a consequence of altered VEGFA levels. Podocyte progenitors also express ephrin B2, another vascular growth factor, and by this might contribute to EphB4 receptor expressing GEnC development and health.¹⁹⁰ Angpt1 is expressed by both podocytes and MCs and binds to the tyrosine-protein kinase receptor (Tie2/Tek) expressed on GEnC. This crosstalk is thought to stabilize the glomerular capillaries, as mice with induced deletion of Angpt1 at embryonic day 10.5 exhibit dilated capillary loops and disrupted subendothelial GBM structures and reduced MCs, whereas podocytes appear intact.¹⁹¹ Podocytes also secrete the chemokine CXCL12 (SDF1), which binds to its receptor CXCR4 on GEnC, a crosstalk important for the formation of glomerular capillaries.²⁹

Despite their importance in glomerular development and maintenance, some podocyte-derived signals have been shown to perpetuate or attenuate GEnC injury in pathologic settings. Therefore, enhanced circulation of endothelin-1 or enhanced podocyte expression of endothelin-1, which binds in a paracrine way to the endothelin receptor A on GEnC, mediates mitochondrial oxidative stress and dysfunction in adjacent GEnC.^{192,193} Furthermore, endothelin 1 induces podocytes to release the GEnC glycocalyx-degrading enzyme heparanase, contributing to GEnC injury in diabetic nephropathy.¹⁹⁴ Another example for disease-perpetuating crosstalk is the podocyte-specific expression of angiopoietin-2, which results in GEnC apoptosis without affecting podocytes.¹⁹⁵ The CXCL12/CXCR4 crosstalk enhances GEnC injury in diabetic nephropathy¹⁷¹ and in Shiga toxin-associated hemolytic uremic syndrome.¹⁷² A GEnC-protective crosstalk with podocytes was suggested for podocyte-derived angiopoietin-like-4 (Angptl4), which is structurally similar to angiopoietins but does not signal over Tie2. Angptl4 was suggested to protect GEnC from oxidative injury in nephrotic syndrome by binding to $\alpha V\beta 5$ integrins.²⁷ A protective reverse signaling from GEnCs to podocytes has been demonstrated for vasohibin secreted from GEnC, which is thought to counteract VEGFA signaling in situations of pathologic elevated VEGFA levels such as in diabetic nephropathy.¹⁹⁶

GLOMERULAR ENDOTHELIAL CELLS AND MESANGIAL CELLS

The fate of MCs and GEnCs is tightly linked. Both communicate directly at the paramesangial areas of glomerular capillaries, where their plasma membranes are in direct contact. Even though MCs secrete a multitude of factors *in vitro*, which could affect GEnC, only a few MCs secreted factors have been identified that partake in GEnC crosstalk in

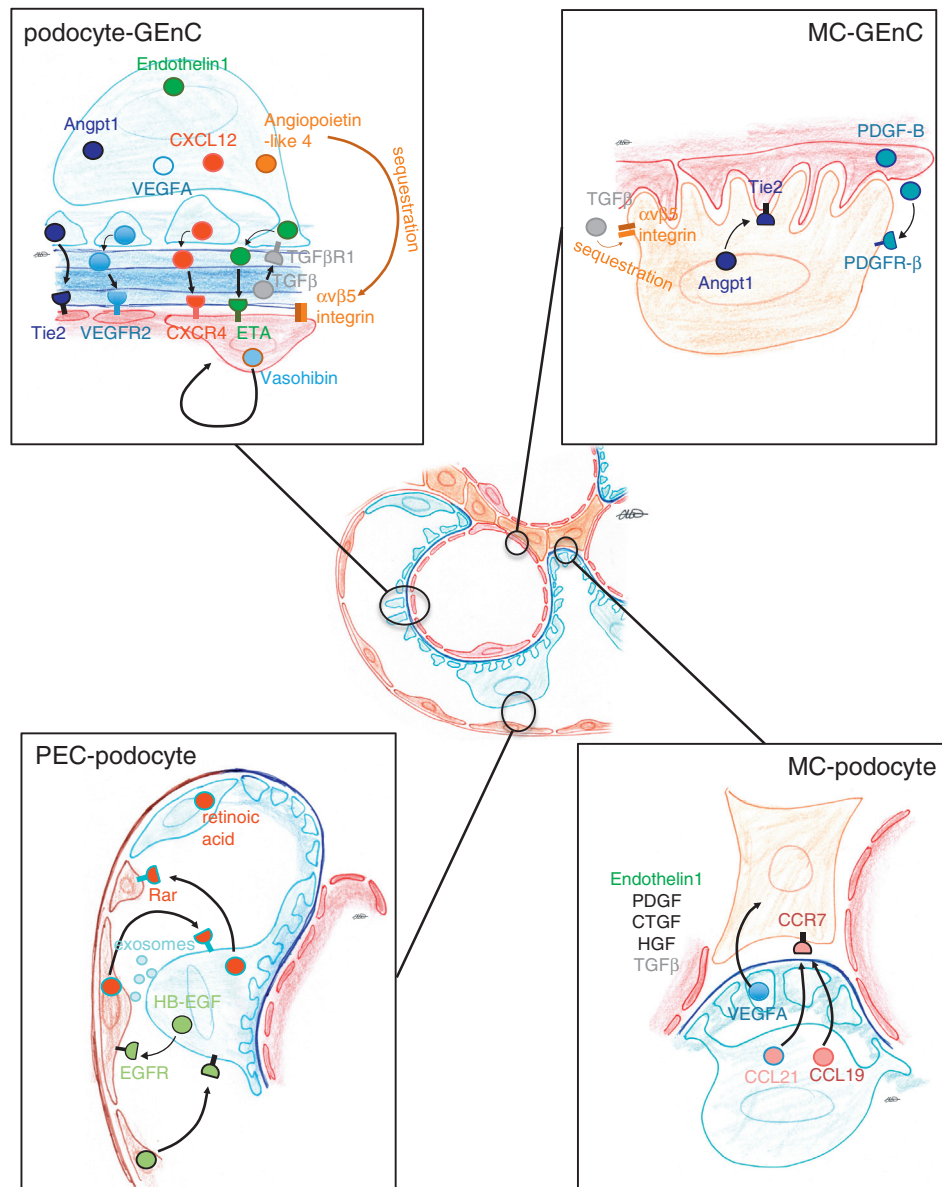


Fig. 4.4 Examples of intra glomerular crosstalk. *Angpt1*, angiotensin 1; *CTGF*, connective tissue growth factor; *CXCL12*, C-X-C chemokine ligand 12; *CXCR4*, chemokine receptor 4; *EGFR*, epidermal growth factor receptor; *Eta*, endothelin-1 receptor A; *HB-EGF*, heparin-binding epidermal growth factor-like growth factor; *HGF*, hepatocyte growth factor; *PDGF*, platelet-derived growth factor; *PDGF-B*, platelet-derived growth factor B; *PDGFRβ*, platelet-derived growth factor receptor β; *Rar*, retinoic acid receptor; *Tie2*, tyrosine-protein kinase receptor 2; *TGFβ*, transforming growth factor β; *TGF-βR1*, transforming growth factor-β receptor 1; *VEGFA*, vascular endothelial growth factor A; *VEGFR2*, vascular endothelial growth factor receptor 2.

vivo. This is in part a consequence of the lack of glomerular endothelial cell and also of MC-specific gene targeting strategies in vivo, which is a prerequisite for such investigations. Endothelial derived PDGF-B and its receptor PDGFR-β, localized on MCs, have been demonstrated to be of crucial importance for the development and maintenance of glomerular capillaries.⁹ Consequently, injury and loss of GENCs—for example, due to toxin- or antibody-related GENC injury—results in decreased PDGF-B levels and MC death (mesangiolysis). MCs maintain endothelial health and function through integrin α5β8-dependent sequestering of TGF-β, thereby reducing the amount of active TGF-β.¹⁹⁷

Furthermore, like podocytes, MCs synthesize angiotensin-1, which binds to its receptor Tie2 on GENC¹⁹¹ to stabilize the vasculature.

PODOCYTES AND MESANGIAL CELLS

It is not clear to date at which sites podocytes communicate with MCs and the in vivo evidence of podocyte–MC crosstalk is scarce. Nonetheless, experimental data and clinical observations in several hereditary forms of nephrotic syndrome due to mutations in podocyte-specific genes¹⁹⁸ suggest that such a communication exists. For example, in glomerular development, mutations in podocyte genes such as the transcription

factor Pod1/tcf21,¹⁹⁹ phospholipase C ϵ ²⁰⁰ laminin $\alpha 5$,⁹² and Wilms tumor antigen⁵⁵ result in a failure of MCs to migrated into glomeruli. The podocyte-specific deletion of collagen type IV $\alpha 3$ (Alport mouse) results in enhanced expression of integrin $\alpha 1$ by MCs,²⁰¹ possibly affecting MCs cell adhesion and cell signaling at the GBM. The chemokines CCL19 and CCL21, generated by podocytes, bind to CCR7 on MCs and are thought to regulate local MC migration and adherence to the GBM.²⁰² Also, a decrease of VEGFA secretion from podocytes results in mesangiolysis,³⁰ supporting the idea of a podocyte-MC crosstalk, which is of relevance in glomerular development. In glomerular injury, experimental data support a communication of podocytes with MCs, although it cannot be excluded that the observed effects on MCs in podocyte injury are not the result from altered PDGF-B levels related to podocyte-dependent GEnC injury. Several signaling pathways might be involved in podocyte-MC crosstalk in the setting of injury, such as endothelin 1, PDGF, CTGF, HGF, and TGF- β .²⁸

PODOCYTES AND PARIETAL EPITHELIAL CELLS

Under physiologic conditions podocytes and PECs are in close proximity at the vascular pole, where transitional cells called parietal podocytes carry both the characteristics of PECs and podocytes.¹⁵⁴ Another theoretical site of cross communication is across Bowman's space, where the apical membranes of podocytes and PECs can overcome the physical separation given by the primary filtrate and touch. Controlled depletion of PECs results in transient proteinuria with focal podocyte foot process effacement,¹⁸¹ and podocytopenia is associated with PEC hyperplasia,²⁰³ suggesting interdependence of both cells. Communication between PECs and podocytes could also ensue from the uptake of podocyte-derived proteins from the primary urine by PECs.²⁰⁴ Furthermore, podocytes release exosomes to the urine^{205,206} that in turn could affect PECs.

In glomerular injury, podocytes are in close contact to PECs by bridges formed between the capillary tuft and Bowman's capsule²⁰⁷ and in intraglomerular crescents, of which they are both constituents.^{176,208} In glomerulonephritis, mathematical multiscale modeling studies²⁰⁹ and experimental data suggest that podocyte and PEC cross communication might regulate proliferation of both cells and regulate regeneration of podocytes from PECs. Proliferation of both cell types and thereby crescent formation was shown to be dependent of the heparin-binding EGF-like growth factor (HB-EGF), which is de novo expressed by podocytes and PECs in RPGN and the EGF receptor, which is found on both cells.²¹⁰ Lineage tracing experiments suggest that podocyte regeneration in glomerulonephritis occurs from renal progenitor cells located in Bowman's capsule⁷⁶ and that this process can be enhanced by retinoic acid.²¹¹ Therefore, retinoic acid synthesized in glomeruli promotes renal progenitor cells to differentiate toward a podocyte phenotype.²¹²

COMMON MECHANISMS OF GLOMERULAR DISEASES

There appear to be several basic responses of the glomerulus to injury such as cellular proliferation, changes in glomerular cell phenotypes, and increased deposition of extracellular

matrix. Any cause of severe glomerulonephritis (GN) can cause a crescent formation (typical for rapid progressive GN), which is composed of parietal cells,^{181,213} podocytes,²⁰⁸ and inflammatory cells.^{214,215} Most forms of glomerular injury result from immunologic mechanisms,^{216–220} which include both humoral and cellular components. Only little is known about the origin of the etiologic agents that induce the immunologic mechanisms, with the exception of infection-related forms of disease such as beta-hemolytic streptococci in poststreptococcal GN, or hepatitis C and hepatitis B virus in cryoglobulinemic membranoproliferative GN. However, it is likely that drugs, toxins, and other infectious agents induce similar immune responses that result in GN via shared common pathways.

The humoral response is often a T helper cell 2 (Th2) mediated response resulting in B-cell activation, antibody generation and deposition, and complement activation. Immunoglobulin and complement component deposition is found in most human glomerular diseases, suggesting that the humoral response is crucial in the development of glomerular injury, which has been the rationale for the use of therapeutic B-cell depletion in various glomerular diseases. There are three patterns of immunoglobulin deposition in the glomerulus: (1) immune deposits at the GBM and in the subepithelial space (underneath podocytes) are typical for membranous nephropathy and usually do not initiate a strong inflammatory reaction, as the deposits are separated from the circulation by the GBM; (2) immune deposits in the subendothelial space (lupus nephritis and membranoproliferative GN); or (3) in the mesangium (IgA nephropathy and lupus nephritis), on the other hand, initiate multiple inflammatory processes. The final pattern of immunoglobulin deposition is determined by the biologic properties of the immunoglobulins (IgG subtype) deposited, the absolute amount of immunoglobulins deposited, and lastly the mechanisms whereby the deposits are formed. Deposition of the complement-fixing IgG1 or IgG3 subtypes ensues in stronger glomerular injury than deposition of IgA or IgG4, which both poorly activate complement.²²¹ Principally, antibody-antigen binding that takes place within the glomerulus to glomerular self or nonself antigens (termed *in situ* binding) induces stronger complement activation than when preformed immune complexes are deposited in glomeruli. Typical glomerular self-antigens are the phospholipase A2 receptor (PLA₂R1)²²² and thrombospondin type-1 domain-containing 7A (THSD7A)²²³ in membranous nephropathy, or the noncollagenous domain of the $\alpha 3$ chain of type IV collagen known as the Goodpasture antigen.^{224,225} Nonself antigens localize to glomerular capillaries by mechanisms such as charge affinity for glomerular structures or pure passive trapping in the glomerular sieve in the form of the antigen alone (termed as *planted* antigens) or as antigen-antibody complexes formed outside the kidney. *In situ* antibody binding to *planted* nonself antigens is typical in lupus nephritis to DNA nucleosome complexes^{226,227} or in IgA nephropathy to abnormally glycosylated IgA.²²⁸ Glomerular deposition of preformed immune complexes to nonself antigens has been demonstrated in early childhood MN, where immune complexes containing cationic bovine serum albumin are deposited,²²⁹ or in hepatitis C virus-associated membranoproliferative GN where hepatitis C virus-containing cryoglobulins are deposited.²³⁰ There is little experimental evidence that immunoglobulin

binding alone induces significant tissue injury, except for when the antibodies bind to podocyte antigens such as nephrin^{231,232} of the slit membrane or to THSD7A.²³³ Of note, severe inflammation can occur with only little antibody deposited, as in antineutrophil cytoplasmic antibody (ANCA)-associated GN.

The cellular response is a largely T helper cell 1 (Th1)-mediated response characterized by the infiltration of circulating mononuclear cells such as lymphocytes and macrophages into glomeruli and the formation of crescents. Neutrophils are the earliest cells to be found in inflamed glomeruli in human biopsies and are strong inducers of glomerular injury.²³⁴ Animal studies demonstrate that their strongest attractants to inflamed glomeruli are interleukin (IL) 8 and complement factor C5a,²³⁵ bound to the glomerular endothelium via HS proteoglycans.^{150,236} Neutrophils are activated by phagocytosis of immune complex aggregates, which induces them to undergo a respiratory burst with generation of reactive oxygen species such as hydrogen peroxide. Hydrogen peroxide interacts with the neutrophil cationic enzyme myeloperoxidase (MPO) to halogenate the glomerular capillary wall.²³⁷ Furthermore, neutrophils store other cationic enzymes such as proteinase 3 (PR3), elastase, and cathepsin G, which upon release further degrade the glomerular capillary wall. Lastly, neutrophils release extracellular traps, web-like DNA structures expelled from nuclei with adherent histones, proteases, peptides, and enzymes, which show a modest contribution to glomerular injury in anti-GBM glomerulonephritis,²³⁸ but could be more injurious in ANCA-associated GN and lupus nephritis.²³⁹

Macrophages are typically found in glomerular lesions with crescents and serve as effector cells of both humoral and cell-mediated forms of immune glomerular injury, because their localization to inflamed glomeruli is induced by interactions with immunoglobulins through their Fc receptors and through chemokines, such as CCL5, and CCL2. Similar to neutrophils, macrophages generate direct tissue injury by the release of oxidants and proteases. Additionally, they release tissue factor to induce glomerular fibrin deposition and crescent formation and TGF- β to induce the synthesis of the extracellular matrix, culminating in glomerular sclerosis.²⁴⁰

T cells are rarely found in injured glomeruli except for GNs primarily mediated by macrophages such as crescentic GN. T-cell-mediated glomerular injury is mostly the result from released chemokines and the recruitment of macrophages.²⁴¹ However, ovalbumin-specific CD4+ and CD8+ T cells together can induce glomerular injury in transgenic mice that express the antigen ovalbumin in podocytes.²⁰⁴ Among the known T-cell subtypes, there is strong experimental evidence for the importance of T helper cell 17 (Th17) in crescentic GN.^{242,243} Th17 cells produce and secrete IL-17A, IL-17F, IL-21, IL-22, which promotes inflammation by directly causing tissue injury and enhancing secretion of proinflammatory cytokines and chemokines by resident cells. This results in augmented infiltration of leukocytes, in particular neutrophils, recruited by CXCL5²⁴⁴ to the affected kidney where they induce further inflammation and injury.²⁴⁵ The kidney-infiltrating Th17 cells are partly recruited from the gut.²⁴⁶

Platelets are present in glomerular lesions in which intracapillary thrombosis is involved, typically observed in thrombotic microangiopathies and antiphospholipid syndrome. Platelets

are important players in the formation of thrombi and in the recruitment of leukocytes to the inflamed glomerulus.²⁴⁷ In addition, they release factors that enhance glomerular permeability to proteins, enhance immune complex deposition,^{248–250} and induce MC proliferation (PDGF)²⁵¹ and MC sclerosis (TGF- β).²⁵²

Dendritic cells (DCs) are restricted to the tubulointerstitium and are absent from glomeruli.²⁵³ However, proteins that pass the glomerular filter are captured by renal DCs or reach the renal lymph nodes by lymphatic drainage²⁵⁴ to induce immune tolerance²⁵⁵ to innocuous proteins, such as food antigens or hormones, or to stimulate infiltrating T cells to produce proinflammatory cytokines.²⁰⁴

The site of glomerular injury, especially when glomerular cell is involved, determines whether the patient has an inflammatory or a noninflammatory injury (Fig. 4.5). Because glomerular endothelial and MCs are in contact with circulating factors such as complement and inflammatory cells, they are prone to react to injury via a principally more dramatic inflammatory response. In contrast, PECs and podocytes are separated by the GBM from the circulation; thus podocyte injury is rarely associated with activation of circulating inflammatory cells. Clinically the distinction between inflammatory and noninflammatory injury is crucial for the adequate diagnosis and management of patients. The clinical characteristics of inflammatory injury are hematuria with dysmorphic erythrocytes with or without red blood cell casts and occasional leukocyturia. Inflammatory injury is accompanied by varying degrees of proteinuria, which ranges from mild to nephrotic range proteinuria and a normal

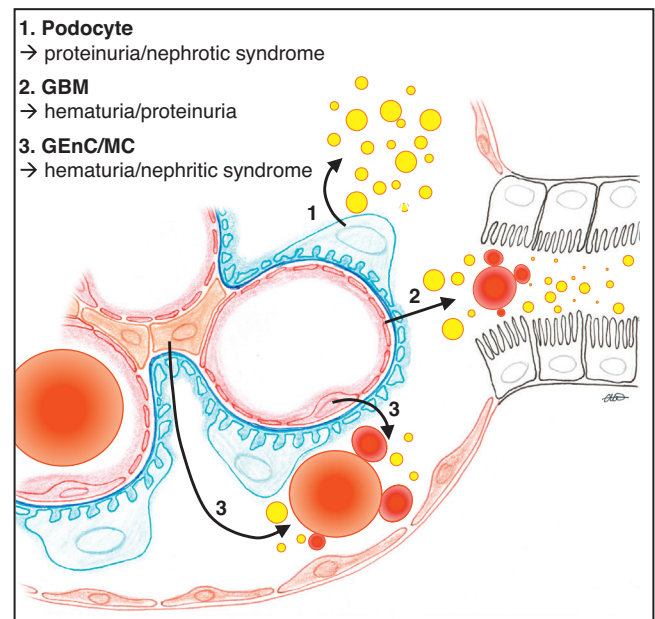


Fig. 4.5 The clinical presentation reflects the localization of glomerular injury. Injury of podocytes results in proteinuria (yellow droplets) eventually leading to the clinical manifestation of nephrotic syndrome. Injury of the glomerular basement membrane (GBM) commonly results in proteinuria and hematuria with dysmorphic erythrocytes. Injury of glomerular endothelial cells (GEnC) and mesangial cells (MC) usually leads to hematuria with dysmorphic erythrocytes and only little proteinuria.

or reduced glomerular filtration rate, depending on the severity of disease. Morphologically, inflammatory injury is characterized by glomerular hypercellularity that results from proliferating resident glomerular (mostly MCs and PECs) and from infiltrating hematopoietic cells (mostly neutrophils and macrophages), phenotype change, and visible structural injury. Glomerular injury arises from the release of inflammatory substances from infiltrating hematopoietic cells and from glomerular cells or from the impairment of protective mediators such as complement factor H²⁵⁶ or complement factor H-related protein 5²⁵⁷ as negative regulators of the complement pathway. The release of inflammatory substances, such as cytokines, growth factors, proteases, products resulting from complement activation (C5a, C5b-9), vasoactive agents, and oxidants,^{217,258,259} initiates thrombosis, necrosis, and crescent formation which, if extensive, leads to the serious clinical condition of rapid progressive glomerulonephritis. Noninflammatory lesions usually involve podocytes and are termed podocytopathies. They are characterized by proteinuria and (if proteinuria is extensive) nephrotic syndrome (a triad consisting of proteinuria over 3.5 g/day, edema, and hypertriglyceridemia) without hematuria.

Clinical Relevance

The site of glomerular injury determines the clinical picture. Whereas involvement of podocytes presents as glomerular injury with proteinuria or nephrotic syndrome, involvement of glomerular endothelial cells, the GBM, and/or MCs typically presents with microhematuria or nephritic syndrome.

MECHANISMS OF INJURY IN COMMON PODOCYTOPATHIES

The most common causes of noninflammatory immune-mediated glomerular injury are minimal change disease (MCD), primary focal segmental glomerulosclerosis (FSGS), and membranous nephropathy (MN). All three entities exhibit a dramatic increase in glomerular permeability with little to significant structural abnormalities by light microscopy in common. These diseases are classified as podocytopathies, as podocytes are thought to be the primary glomerular cell affected in the pathogenesis. In contrast, “nontraditional” podocyte diseases include diabetic kidney disease, human immunodeficiency virus nephropathy, amyloidosis, Fabry disease, membranoproliferative glomerulonephritis, and postinfectious glomerulonephritis. Diabetic nephropathy (covered in detail in Chapter 39) is the numerically and economically most important form of progressive kidney disease worldwide, involving the podocyte, endothelial, and MCs with a perturbation of glomerular cell crosstalk. The inciting causes of each podocyte disease differ, and therefore each disease affects podocytes in different ways; in turn, the response to injury in each disease differs, leading to different histologic and clinical manifestations. Yet, regardless of the inciting causes and their mediators, several common clinical and pathologic responses occur in podocyte injury, as highlighted earlier—namely, hypertrophy, foot process effacement, loss, and proteinuria.

MINIMAL CHANGE DISEASE AND FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Characteristics of MCD: In MCD the glomerulus is per definition mostly normal by light microscopy with absence of complement or immunoglobulin deposition. In light of the histologic “minimal changes,” pathologic diagnosis is primarily based on electron microscopic evaluations of podocytes, which exhibit foot process effacement, microvillous transformation, and vacuolization. The absence of glomerular sclerosis differentiates MCD from FSGS. MCD typically presents as a steroid-sensitive nephrotic syndrome (SSNS), contrasting with FSGS, which often presents with steroid-resistant nephrotic syndrome (SRNS). Whether steroid-resistant MCD exists as its own disease entity, or whether steroid resistance in histologic MCD represents an early form of FSGS is a matter of debate, considering the high rate of sampling error on renal biopsies, possibly missing present (but rare) glomeruli with sclerosis. It is likely that patients with MCD can progress to FSGS due to a consistent pathologic agent.²⁶⁰

Characteristics of FSGS: FSGS is a generic term for a histologic injury pattern defined by segmental glomerular consolidation into a scar that affects some but not all glomeruli with a wide range of etiologic interpretations. FSGS describes both a disease characterized by a primary podocyte injury (primary FSGS), and a lesion that occurs secondarily in any type of chronic kidney disease (secondary FSGS).²⁶¹ There is abundant evidence that classical FSGS is the consequence of podocyte loss in experimental models,²⁶² which is accompanied by proliferation and migration of PECs to the glomerular tuft (both discussed earlier in the chapter). The underlying causes or mechanisms of FSGS are broadly considered as hereditary/congenital and sporadic/acquired in nature. Primary FSGS presents either with SSNS if triggered by causative circulating agents that increase the permeability of the glomerular filtration barrier (as discussed later) or with SRNS in case of an underlying genetic cause. Secondary FSGS, on the other hand, is associated with nephron loss, drug toxicity, or viral infections and rarely presents with nephrotic syndrome, but is steroid resistant.

PATHOPHYSIOLOGIC CONCEPTS OF MCD AND PRIMARY FSGS

MCD and primary FSGS might represent different histologic patterns of the same disease entity. At least there is a significant overlap in the factors causing these podocytopathies, which are the following:

1. **Soluble serum factors:** Based on experimental data, soluble serum factors are thought to be causative in MCD and FSGS, as nephrotic plasma has direct cellular effects on cultured podocytes²⁶³ or in single perfused kidneys.²⁶⁴ Many efforts have been undertaken to unravel “the increasing or missing circulating permeability factor” in SSNS and SRNS.²⁶⁵ Roles have been indicated for TNF α ,^{266,267} circulating cardiotrophin-like cytokine factor 1 (member of the IL-6 family), circulating hemopexin,^{268,269} and the soluble urokinase-type plasminogen activator receptor (suPAR)²⁷⁰ in the development of nephrotic syndrome.²⁷¹ These factors, however, need further verification in the context of human disease activity,^{272,273} disease specificity,^{274–276} and therapeutic effects.²⁷⁷

2. **Immune dysfunction:** Considerable clinical and experimental evidence point toward an immune dysfunction on the T- and B-cell side in MCD. Clinically, MCD is not only very responsive to steroids but also to rituximab,^{278,279} a monoclonal antibody against plasma cell CD20, and a significant association of HLA-DQA1 (a major histocompatibility complex class II) missense coding variants exists.²⁸⁰ Additionally, MCD is associated with Hodgkin disease and with allergies.²⁸¹ On the experimental side, mice develop proteinuria if they receive CD34+ peripheral stem cells from MCD patients²⁸² or following the injection of the supernatant derived from T cells or from peripheral blood mononuclear cells.²⁸³ T- and B-cell dysfunction is further suggested by a different DNA methylation pattern of Th0 cells in MCD patients,²⁸⁴ an altered Th17/regulatory T-cell balance,²⁸⁵ and an upregulation of T-cell-derived IL-13 in patients¹⁶⁰ that can cause proteinuria and foot process effacement in rats.^{286,287} Further pointing toward an immune dysfunction as the origin of nephrotic syndrome is the finding that abatacept, an antibody challenging the CD80 (B7-1)–CTLA-4-axis has been shown to reduce proteinuria in FSGS,²⁸⁸ an effect requiring still further confirmation.^{289–291} Recent experimental findings have given rise to the idea of a disease-specific expression of CD80^{288,292} or of the expression of a hyposialylated form of angiopoietin-like-4 by podocytes, which might contribute to glomerular disease.^{293–295}
 3. **Genetic inheritance:** Familial cases are rare in MCD, making genetic causes for MCD unlikely. Nonetheless, whole-exome sequencing in SSNS patients revealed mutations in epithelial membrane protein 2 (EMP2), which is involved in regulating endocytosis and transcytosis,²⁹⁶ shedding new light into possible causes for podocyte disruption in SSNS. There are changes in expression patterns of podocyte-specific transcripts²⁹⁷ and proteins²⁹⁸ in MCD, but it is difficult to determine whether these are cause or effect. There is strong evidence for genetic causes leading to FSGS. Inheritance of genetic causes of primary FSGS may be autosomal dominant or recessive, with a subset of autosomal recessive SRNS presenting as congenital nephrotic syndrome. The biologic functions altered in podocytes by the gene mutations involved in FSGS are broad, ranging from cytoskeletal regulation, slit membrane function, lysosomal function, mitochondrial function, and attachment to the GBM.⁴⁴ In the late 1990s positional cloning of the gene responsible for congenital nephrotic syndrome of the Finnish type led to the identification of the archetypal podocyte-specific protein, nephrin.⁴⁹ This was rapidly followed by the identification of other proteinuric diseases linked to podocyte-specific single-gene disorders, including those affecting podocin,⁵⁰ Wilms tumor 1,²⁹⁹ CD2AP,³⁰⁰ α -actinin-4,⁵¹ TRPC6,³⁰¹ phospholipase C ϵ 1 (PLCE1),³⁰⁰ WW and PDZ domain-containing 2 (MAGI2),³⁰² kidney ankyrin repeat-containing protein (KANK), and others.^{303,304} In each of these conditions it is generally accepted that proteinuria results directly from the disruption of these constitutively expressed genes in the podocyte, leading to FSGS.
- It is becoming increasingly clear that monogenic inheritance of abnormalities in podocyte-specific genes is only one aspect of FSGS. Even though the amount of over 50 known mutations of podocyte genes involved in SRNS

is steadily increasing, these mutations are rare and explain less than 30% of patients with hereditary cases and only 20% of patients with sporadic cases of FSGS.⁴⁴ Studies of the increased susceptibility of African Americans to FSGS have implicated variants of another gene expressed in podocytes, apolipoprotein L1 (APOL1).^{305,306} Experimental expression of the APOL1 risk alleles in a podocyte specific manner demonstrated that these were causal for podocyte foot process effacement, proteinuria, and glomerulosclerosis.³⁰⁷ Mechanistically, the risk-variant APOL1 alleles interfere with endosomal trafficking and block autophagic flux, ultimately leading to inflammatory-mediated podocyte death and glomerular scarring.³⁰⁷ A key question for practicing nephrologists is whether podocyte-specific gene mutations or polymorphisms play a role as predisposing factors for the much more common “sporadic” forms of proteinuric disease. In human disease an opportunity to study the very earliest features of FSGS is afforded by studies of FSGS recurrence in transplanted kidneys; changes in podocytes can be seen in reperfusion biopsies and are predictive of full-blown FSGS recurrence.³⁰⁸ Chapter 43 includes an in-depth discussion of inherited glomerular diseases.

4. **Altered posttranslational regulation of podocyte proteins:** Besides abnormalities in podocyte genes, altered post-translational regulation of podocyte mRNA by small noncoding RNA molecules, termed microRNAs, have been shown to result in FSGS, such as the transcription factor Wilms tumor protein 1 (WT1) by microRNA-193a¹⁶¹ or of Notch1 and p53 by microRNA-30.³⁰⁹

MEMBRANOUS NEPHROPATHY

Membranous nephropathy is an autoimmune disease with the morphologic hallmarks of GBM thickening, granular staining for human IgG and complement components along the glomerular filtration barrier, and subepithelial (subpodocyte) electron-dense deposits by electron microscopy. Although this histologic pattern can arise from the deposition of preformed immune complexes as in lupus nephritis type 5 or from subepithelial antigen trapping in hepatitis B virus and hepatitis C virus-associated secondary forms of MN, primary MN is thought to be the consequence of in situ binding of autoantibodies to podocyte-expressed antigens. The concept that primary MN is an antibody-mediated disease has been supported by the discoveries of autoantibodies to podocyte membrane antigens such as neutral endopeptidase (NEP),³¹⁰ the phospholipase A₂ receptor (PLA₂R1),²²² and thrombospondin type 1 domain-containing 7A (THSD7A).²²³ PLA₂R1 and THSD7A are targets for a malfunctioning immune system in 70% and 5% of adult cases, respectively, and NEP is important in a small number of neonates with MN caused by alloimmunization due to vertical transfer of antibodies from a genetically NEP-deficient mother.²²¹ PLA₂R1 polymorphisms influence the susceptibility to MN, and the association between certain HLA-DQA1 alleles and MN suggests that these HLA class II molecules could facilitate autoimmunity against PLA₂R1.³¹¹ In contrast to PLA₂R1-associated MN, THSD7A-associated MN has a high coincidence of malignancy.^{312,313} The direct pathogenicity of autoantibodies is suggested by the observations that PLA₂R1 or THSD7A

autoantibodies are present in patients with a rapid recurrence of MN in renal transplants,^{223,314,315} the finding that anti-PLA₂R1 antibody levels are associated with disease remission^{316,317} and progression,^{317–319} and the finding that MN can be induced in mice that normally express THSD7A on podocytes^{320,321} by injection of human autoantibodies²³³ or rabbit antibodies to THSD7A.³²² So far there is no proof that PLA₂R1-specific autoantibodies are pathogenic, related to the fact that rodents necessary for such studies normally do not express PLA₂R1 on podocytes. The role of complement in the pathogenesis of human MN remains unclear. First, the deposited IgG in idiopathic MN is typically of the noncomplement fixing IgG4 subtype. Second, in rodent models of MN, clinical and morphologic MN can be induced in the absence of detectable complement deposition,^{233,322} as well as in rodents with genetic deficiency in complement components.^{323,324} On the other hand, C3 and the membrane attack complex C5b9 are usually constituents of the deposits³²⁵; C5b9 inserts into the podocyte membrane and is transported across the cell³²⁶ and excreted into the urine, where high levels of C5b9 can be measured in humans.³²⁷ Experimental studies allow for an injurious as well as a protective function of C5b9. Sublytic C5b9 has been shown to induce podocyte injury by multiple pathways such as the activation of kinases, the induction of endoplasmic reticulum stress, and the production of extracellular matrix.³²⁸ On the other hand, C5b9 enhances the ubiquitin proteasome system,³²⁹ which supports a protective removal of damaged proteins.

EFFECTS OF EXISTING THERAPIES ON PODOCYTES

The basis for therapy of primary nephrotic syndrome (reviewed in Chapter 33) is mostly of a supportive nature, including antihypertensive and antiproteinuric therapy and dietary recommendations.³³⁰ Regrettably, no therapeutic approaches are currently available that specifically target podocytes in disease. However, several therapies have, in addition to their systemic effects, direct biologic actions on podocytes (i.e., pleiotropic actions), and these will be considered in the following sections.

RENIN ANGIOTENSIN SYSTEM BLOCKADE

Blockade of the renin angiotensin system belongs to the standard supportive therapy regimen for primary podocytopathies with proteinuria and remains unchallenged. Systemic and glomerular overactivation of the renin angiotensin system (RAS) through its main effector angiotensin II is central to the pathogenesis of proteinuric glomerular diseases; levels of tissue angiotensin II and the angiotensin subtype 1 receptor are increased in glomeruli³³¹ and podocytes³³² in primary podocytopathies. Activation of RAS is detrimental to glomerular cells including podocytes as it promotes multiple trophic effects, such as apoptosis, ECM protein accumulation, reactive oxygen species production, oxidative stress, alteration in slit diaphragm proteins partly by epigenetic modulation of nephrin promoter methylation,³³³ increased calcium influx through TRPC6 channels,^{334,335} cell cycle inhibition, detachment, and inflammatory cytokine production.^{332,336} Blocking the RAS with angiotensin-converting enzyme inhibitors, angiotensin 1 receptor (AT1R) antagonists, and

mineralocorticoid receptor blockers reduces proteinuria resulting in renoprotection, an effect that is attributed to a reduction in glomerular hydrostatic pressure and an abolishment of the detrimental trophic glomerular effects mentioned earlier.³² For these reasons, inhibition of the renin angiotensin aldosterone system is currently the standard of care for lowering proteinuria.

GLUCOCORTICOIDS

Steroids are immunomodulatory drugs widely used in the treatment of proteinuric diseases, but their modes of action, especially in the noninflammatory forms of nephrotic syndrome such as MCD and primary FSGS, remain completely unknown. Glucocorticoid receptor (GR) expression is ubiquitous; therefore, these drugs could affect any glomerular cell type. It has been shown that GR-induced signaling pathways are functional in murine podocytes, having transcriptional and posttranscriptional effects on podocyte genes.³³⁷ Initial reports in murine³³⁸ and human³³⁹ podocytes showed that dexamethasone had potent biologic effects directly on podocyte structure and function. These include limiting podocyte apoptosis,³⁴⁰ induction of podocyte differentiation by restoring the actin cytoskeleton,³³⁸ increasing levels of the transcription factors Kruppel-like factor 15 (KLF15),³⁴¹ and by preventing the downregulation of protective microRNA-30.³⁰⁹ Specifically, the slit membrane protein nephrin is affected by steroids, as steroids enhance the transport of nephrin from the ER³⁴² and induce the phosphorylation of nephrin,³⁴³ which is reduced in MCD³⁴⁴ and important for the function of nephrin and therefore of the slit membrane.

EVIDENCE FOR DIRECT ACTIONS OF IMMUNOSUPPRESSANTS ON PODOCYTES

The calcineurin inhibitors cyclosporine and tacrolimus are widely used in nephrotic syndrome, either alone or in combination with other therapies. Calcineurin is a Ca²⁺-dependent phosphatase, which dephosphorylates nuclear factor of activated T cells (NFAT), a transcription factor. Dephosphorylation of NFAT initiates its cytoplasmic to nuclear translocation, resulting in an increased transcription of genes such as TRCP6,³⁴⁵ whose gain of function mutations in podocytes induce FSGS.³⁰¹ In accordance, podocyte-expressed NFAT is a strong inducer of glomerulosclerosis in mice.³⁴⁶ Besides their known immunomodulatory effects in T cells, calcineurin inhibitors affect the podocyte cytoskeleton by transcriptional downregulation of the earlier mentioned calcium channel TRPC6,³⁴⁷ and by preventing the degradation of the actin-organizing protein synaptopodin³⁴⁸ in an NFAT-independent manner. Together, these mechanisms appear to result in a stabilization of the podocyte actin cytoskeleton and direct reduction of proteinuria.

The specific anti-B-cell monoclonal antibody rituximab, increasingly thought to be effective in proteinuric diseases even when they are not all obviously immune mediated,³⁴⁹ has been shown to have direct effects on podocytes, including stabilizing their actin cytoskeleton.³⁵⁰ Although monoclonal antibodies are assumed to have very specific binding targets, they can also have “off-target” effects. In this case it seems that rituximab, as well as binding to the CD20 molecule that is its accepted molecular target, also binds to a podocyte protein called sphingomyelin phosphodiesterase acid-like-3b (SMPDL-3b) and this protein stabilizes the actin cytoskeleton.³⁵⁰

IDENTIFICATION OF CANDIDATE THERAPEUTIC APPROACHES FOR THE FUTURE

End-stage renal disease constitutes an enormous burden of morbidity, both to patients who suffer a lifelong chronic disease and to health services that are challenged by high costs arising from dialysis and transplantation. Despite major advances over the past 15 years in unraveling genetic and biologic causes of podocyte dysfunction as the origin of most forms of nephrotic syndrome, glomerular target cell-directed therapies are still in their infancy. Especially in comparison to other specialties in medicine, only a few new drugs have been approved for renal failure in comparison to many new drugs for cancer therapy or for heart diseases. Nonetheless, the separation of monogenic causes of podocyte dysfunction from other causes is now used to stratify those patients in whom immunosuppression can be minimized due to low potential of success. Furthermore, the discovery of the genes involved in hereditary proteinuric disease unraveled critical (and in the future potentially drug able) pathways required for podocyte maintenance, for stabilization of the podocyte actin cytoskeleton and slit diaphragm, and for restoration of metabolic and mitochondrial function.³⁵¹ Several experimental approaches of recent years have aimed at finding podocyte-specific interventions, which stabilize the actin cytoskeletal backbone of podocyte foot processes and by this also of the slit diaphragm.^{352–355} Targeting the metabolic and mitochondrial function in podocytes seems to be another therapeutic option for many forms of podocyte injury. Even though mTOR inhibition is known to induce proteinuria as a typical side effect,^{356,357} inhibition of an overactivated mTOR pathway in the setting of metabolic super fluency, as in diabetic nephropathy, is beneficial for podocyte function and proteinuria.⁶¹ Other examples with therapeutic potential targeting metabolic pathways include lipid lowering (e.g., statins),^{358,359} antidiabetic agents such as thiazolidinediones (TZDs), or glitazones that activate the peroxisome proliferator-activated receptor γ (PPAR γ),^{360–363} and manipulation of VEGF levels³⁶⁴ or autophagy.³⁶⁵

SUMMARY

Kidney diseases with glomerular involvement account for the vast majority of end-stage renal diseases. In the past two decades much has been learned about the glomerulus, a functional and integrated syncytium of four types of glomerular cells, which together ascertain glomerular filtration. Podocytes and GEnC comprise the glomerular filtration barrier. Together, both cells allow for a size- and charge-selective glomerular filtration due to their specialized three-dimensional structure, extensive glycocalyx coating, and synthesis of the unique GBM. MCs regulate glomerular filtration by means of contraction and release of vasoactive substances and maintain the health of glomerular endothelial cells. PECs, built Bowman's capsule to prevent leakage of the primary urine to the tubulointerstitium, can contribute to glomerular scarring, and are thought to constitute a potential reservoir for podocytes in development, maturation, and eventually in adulthood. A cell type-specific view of

glomerular physiology and pathophysiology has enhanced our understanding of glomerular cell biology immensely in the past decades. However, the intricate interactions of glomerular cell types are getting more and more center stage attention as clinical and experimental observations demonstrate that the normal functioning of glomerular filtration requires a coordinated interaction of all four cell types and that injury of one glomerular cell type usually affects the others. Several clinical and experimental challenges and opportunities lie ahead. Identification, designing, and delivering glomerular cell specific therapeutic agents is actively being pursued, both to enhance efficacy and to reduce systemic side effects. Noninvasive diagnostic testing is being keenly studied, such as measuring glomerular cell products in the urine, and markers in the serum and urine, which will hopefully translate into clinical practice. The past two decades have witnessed phenomenal advances in understanding glomerular cell biology in health and disease, which hopefully soon translates into better therapeutic options for progressive glomerular and renal kidney disease.

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Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

- Which glomerular structure is not involved in regulating glomerular filtration?
 - Glomerular endothelial cells
 - Parietal epithelial cells
 - Podocytes
 - Mesangial cells
 - Glomerular basement membrane

Answer: b

Rationale: Glomerular filtration is regulated by the coordinate interaction of glomerular endothelial cells, podocytes, and mesangial cells. Podocytes and glomerular endothelial cells comprise the glomerular filtration barrier. Together, both cells allow for a size- and charge-selective glomerular filtration due to their specialized three-dimensional structure, extensive glycocalyx coating, and synthesis of the unique glomerular basement membrane. Mesangial cells regulate glomerular filtration by means of contraction and release of vasoactive substances and maintain the health of glomerular endothelial cells. Parietal epithelial cells built the Bowman's capsule to prevent leakage of the primary urine to the tubulointerstitium and are not primarily involved in glomerular filtration. They can, however, contribute to glomerular scarring and are thought to constitute a potential reservoir for podocytes in development, maturation, and eventually in adulthood.

- A 26-year-old female patient presents with swollen legs for the past week. She has been using ibuprofen 800 mg 3× a day for 1 month for a shoulder injury. Her examination is significant for a blood pressure of 130/90 mm Hg and 3+ lower extremity edema. Urinalysis: exhibits 4+ protein, no blood, 24-h urine protein: 10.8 g, no hematuria, Serum creatinine 0.9 mg/dL (eGFR >60 mL/min).

What is the most likely glomerular location causing this clinical picture?

- Mesangial cells (i.e., IgA nephritis)
- Glomerular basement membrane (i.e., Alport syndrome)
- Glomerular endothelial cell (i.e., eclampsia)
- Parietal epithelial cell (i.e., rapid progressive glomerulonephritis)
- Podocyte (i.e., minimal change disease)

Answer: e

Rationale: The predominant symptom of massive proteinuria without hematuria in the setting of preserved renal function is characteristic of primary podocyte injury such as in minimal change disease. In this case, minimal change disease might be a result of the use of nonsteroidal antiinflammatory drugs. Mesangial cell, glomerular basement membrane, and glomerular endothelial cell involvement usually presents with mild to strong hematuria with characteristic dysmorphic erythrocytes. Parietal epithelial cell involvement is reflected by the formation of glomerular scars.

- Which of the following are podocyte antigens in adult membranous nephropathy?
 - suPAR
 - Neutral endopeptidase (NEP)
 - Phospholipase A2 receptor (PLA₂R1)
 - Nephrin
 - Thrombospondin type-1 domain-containing 7A (THSD7A)

Answers: c and d

Rationale: NEP, PLA₂R, and THSD7A are all podocyte antigens associated with membranous nephropathy, however, NEP is only found in rare cases of antenatal membranous nephropathy caused by alloimmunization due to vertical transfer of antibodies from a genetically NEP-deficient mother. Autoantibodies to PLA₂R1 and THSD7A are typically found in adult membranous nephropathy. PLA₂R1 is highly specific for membranous nephropathy and is found in 70% of patients with membranous nephropathy and is associated with disease activity. Autoantibodies to THSD7A are found in 5% of patients with membranous nephropathy and are strongly associated with the tumors.